March 14, 2025

To Whom It May Concern,

On February 6, 2025, on behalf of the United States Office of Science and Technology Policy (OSTP), the Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO) requested input on the Development of an Artificial Intelligence (AI) Action Plan to define the priority policy actions needed to sustain and enhance America's AI dominance, and to ensure that unnecessarily burdensome requirements do not hamper private sector AI innovation.

Introduction

LineVision is pleased to submit these comments to provide input on the highest priority policy actions that should be reflected in the new AI Action Plan. LineVision is a U.S.-based Grid-Enhancing Technology (GETs) company that provides electric utilities with monitoring solutions for high-voltage transmission lines which can unlock as much as 40% additional capacity on existing lines through Dynamic Line Ratings (DLR). LineVision's non-contact sensors and sophisticated analytics also enable actionable insights into the real-time status and long-term health of transmission lines while improving situational awareness, helping to ensure optimal, safe, and reliable operation.

DLR is a key part of AI infrastructure: a critical, fuel-agnostic technology to help the nation's grid keep up with the extreme load growth that will be needed to ensure US energy and AI dominance. Readily deployable in mere months, DLR will be a key tool for getting data centers connected to the grid more quickly, as the grid is already capacity-constrained throughout much of the U.S.

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Background

The U.S. houses a vast number of data centers, which feed power-hungry AI applications, helping keep the U.S. the clear dominating force in AI development. These data centers are driving massive load growth where they are being developed and have already begun to strain portions of the power grid. The ability for data centers to connect to the power grid is directly tied to the availability of capacity on the grid. This capacity, however, is quickly being exhausted.

If the U.S. wishes to maintain its role as the world leader in AI, the power grid cannot be a barrier to the deployment of the key infrastructure that is needed to power AI growth - data centers. The advancement of AI will only occur if there are sufficient data centers to support its growth. The grid is the infrastructure that powers these critical facilities, and if its capacity is not quickly expanded, America is at risk of falling behind in the AI race.

<u>The Challenge - Data Centers Cannot Connect to the Grid Quickly Enough</u>

In October 2023, the U.S. Department of Energy published its National Transmission Needs Study, which indicated that interregional transfer capacity would need to increase by 412% by 2035 in order to accommodate a high load growth future.[1] Since then, forecasts for data centers and other large industrial loads have only increased.

Beyond the energy supply needed to provide power to data centers, grid capacity must be available to connect these data centers to the grid in the first place. The Center for Strategic and International Studies (CSIS) refers to this problem in terms of "speed-to-power" - the measure of how fast a potential data center site can access the electricity needed to power its stock of chips.[2] As highlighted by CSIS, speed-to-power far outweighs any other factor in slowing the connection of new data centers: "Access to electricity supply is the binding constraint on expanded computational capacity and therefore on continued U.S. leadership in AI." GETs like DLR are a key tool to solving the speed-to-power challenge on an already-constrained power grid.

There are three ways to materially expand transmission capacity: 1) get more capacity out of existing lines through GETs; 2) rebuild/replace existing lines with newer technology (i.e., high performance conductors) with higher capacity levels, also known as re-conductoring; and 3) build new transmission lines. Building new transmission lines, on average, takes 7-10 years. Re-conductoring typically takes 18-24 months to complete. But deploying GETs to expand the capacity of the existing grid - such as Dynamic Line Ratings (DLR) - takes as little as three months.

For the U.S. to maintain its role as the world leader in AI, a portfolio approach is needed, meaning some of all of the above. New transmission capacity - whether through newly-built infrastructure or simply getting more out of existing infrastructure using GETs - will be necessary to serve this significant new load associated with data centers and AI.

[1] https://www.energy.gov/sites/default/files/2023-12/National%20Transmission%20Needs%20Study%20-%20Final_2023.12.1.pdf
[2] https://www.csis.org/analysis/electricity-supply-bottleneck-us-ai-dominance

Data centers can be built and connected to the grid in a fraction of the amount of time it takes to build new transmission infrastructure (usually 7-10 years). According to McKinsey, in 2022, total U.S. data center demand was 17 GW[3], which of course has only grown since then, but not at the clip experts are now seeing. Between now and 2030, utility estimates show data center load growth that may exceed 90 GW.[4] In other words, over the next six years the U.S. could build more than five times the capacity of data centers than existed in the entire country just three years ago. AI is expected to represent 70% of that total growth in data center demand.[5] This level of demand growth will need not only new energy generation, but also new transmission capacity to handle it.

In order to connect to the grid and operate, data centers need transmission capacity. Unique as compared to other loads, like residential homes or 9-to-5 businesses, data centers generally operate 24/7. These high load factor facilities are already putting a strain on the grid - a problem that will only become more severe if transmission capacity is not added in the near-term. According to Goldman Sachs, "data center supply — specifically the rate at which incremental supply is built — has been constrained over the past 18 months." These constraints have arisen from the inability of utilities to expand transmission capacity because of permitting delays, supply chain bottlenecks, and infrastructure that is both costly and time-intensive to upgrade.[6]

With new data centers being built at a faster clip than ever, these constraints will only get worse. And with newly-built transmission usually taking 7-10 years to plan, site, permit, construct, and begin operation, even if we were to cut that time in half, the U.S. will begin to lag in its ability to connect data centers. Using technology solutions like DLR to expand the grid's capacity needs to be part of the AI Action Plan, whether through new incentives, requirements, or otherwise.

<u>The Solution - DLR is Needed to Get More Out of the Existing Grid</u>

A key factor in the availability of transmission capacity is based on the rating of a transmission line, which determines how much power any given line can safely pass through. Historically, these ratings have used static measurements based on conservative assumptions to measure that capacity. But in reality, the ability for a transmission line to carry less or more capacity is based largely on how hot or cool

[3] https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/investing-in-the-rising-data-center-economy
[4] https://gridstrategiesllc.com/wp-content/uploads/National-Load-Growth-Report-2024.pdf
[5] https://gridstrategiesllc.com/wp-content/uploads/National-Load-Growth-Report-2024.pdf
[6] https://www.goldmansachs.com/insights/articles/ai-to-drive-165-increase-in-data-center-power-demand-by-2030

that line is. DLRs measure ambient conditions, such as heating and wind, that heat up or cool down transmission lines. DLRs calculate the real-time and forecasted capacity of transmission lines by factoring in real-world conditions, removing the historical approach of relying on those conservative assumptions. For example, DLR can identify times when there is additional ampacity available on a line, meaning the ability to safely pass through more power. It can also identify times when the line's true carrying capacity has been overstated, posing an operational risk.

DLR has already shown the ability to bolster transmission capacity by as much as 40% - a huge jump, that, if operationalized, would allow the grid to accommodate increased amounts of data center load that it otherwise cannot accommodate today. In addition to being deployed on existing lines to get more capacity out of today's grid, GETs like DLR can be a component of newly-built transmission, as it has shown the ability to increase the utilization of new/upgraded transmission by an average of 16%.[7]

DLR can be deployed and sending data to utilities in mere months, quickly allowing them to determine where on the grid they can increase transmission capacity, in turn allowing them to more quickly get more loads connected to the grid. Recognizing the massive costs and time associated with building new transmission, the fastest option to ensure the grid's readiness to serve AI-fueled data centers will be through deploying DLR. New transmission lines, along with re-conductoring, will certainly be other key tools in the toolkit as well, but the low cost and quick ability of DLR to be deployed will give the US the grid capacity it needs to maintain its global leadership role in AI development.

While DLR has been able to demonstrate significant grid benefits, utilities today lack a direct requirement or incentive to deploy it. Without policy levers to either require utilities to deploy DLR where cost-effective, or incentivize them to do so, the build-out of DLR across the US grid risks being too slow to impact the near-term need to connect the data centers needed to continue AI development.

According to US research institute RAND, China has already taken a significant lead over the US in its ability to harness transmission technologies.[8] China is already the world leader in building transmission, raising national security concerns, and allowing Chinese companies to grow their global market share. The US will need to expedite

[7] Building a Better Grid: How Grid-Enhancing Technologies Complement Transmission Buildouts. Available at: https://www.brattle.com/wp-content/uploads/2023/04/Building-a-Better-Grid-How-Grid-Enhancing-Technologies-Complement-Transmission-Buildouts.pdf

[8] https://www.rand.org/pubs/commentary/2024/01/the-us-must-close-the-long-distance-power-transmission.html

the deployment of critical AI infrastructure like DLR in order to compete globally in the AI race, and policy action is necessary to doing so.

<u>Policy Action is Necessary to Ensuring DLR is Deployed Sufficiently to Accommodate AI Growth</u>

Despite the variability of forecasts, load growth from data centers is expected to continue increasing in the coming decade. While it can be difficult to forecast exactly how much load will come online and when, being proactive and preparing to meet this data center load growth in a cost-effective manner is critical to further enhancing America's AI dominance in a way that maintains the grid's safety and reliability.

A lack of timely rules and regulations around GETs will slow their deployment. Policymakers have a vital role to play in ensuring that evolving regulation keeps up with evolving technology. The Federal Energy Regulatory Commission's (FERC) inclusion of DLR in Order 1920-A and its 2024 Advanced Notice of Proposed Rulemaking (ANOPR) on DLR[9] underscore the federal government's recognition of DLR's potential to cost-effectively solve ever-increasing grid constraints.

Federal and state entities have a number of priorities to balance when it comes to the growth of data centers and large electrified loads in general. These priorities include fostering an environment to support economic growth, keeping energy costs affordable for all customers, and maintaining a safe and reliable transmission and distribution grid. Potential action could include:

- **FERC:**
  - Advance the DLR ANOPR to a NOPR, expedite the implementation timeline so that it can allow the US to be AI-dominant; consider a requirement to deploy DLR not just in grid-congested areas, but in states where data centers are seeking to connect large loads to the grid.
  - Ensure that FERC stays on top of Order 1920-A compliance with regard to DLR evaluation and manages a reasonable implementation of the rule.

- **DOE:**
  - Fund research, pilots, studies on planning frameworks that accommodate data center growth like the Collaborations Advancing Rapid Load Additions (CARLA) program, third-party procurement frameworks, best ways for DLR to serve data centers aligning lack of firm, fixed capacity with flexible data center energy management approaches.

[9] See FERC Fact Sheet here: https://www.ferc.gov/news-events/news/ferc-seeks-comment-potential-dlr-framework-improve-grid-operations-fact-sheet

- Provide loans and grants (such as the GRIP grants) to utilities who deploy DLR to accommodate and expedite data center growth.

Conclusion

If the U.S. is to maintain global leadership in AI, it can't do so without data centers, which in turn can't be built fast enough unless DLR is able to expand the capacity of the existing grid. This will necessitate the use of every tool in the toolbox. Given the challenges in siting, permitting and constructing transmission lines – a process that can take a decade or more – meeting this demand in the near-term will require the timely use of commercially available technologies, such as DLR, which can unlock capacity on the existing grid in a matter of months.

Steps taken now to meet the demand from data centers powering AI innovation by deploying GETs like DLR will yield significant benefits and should be a critical part of the new AI Action Plan.

Sincerely
LineVision, Inc